

Monsij Biswal

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EDUCATION

MS/PhD, Electrical and Computer Engineering, University of California, Santa Barbara 2020-Present

- *Advisor*: Prof. Kenneth Rose
- *Major*: Communications *Minor*: Signal Processing *Award*: Departmental Funding
- *Relevant Courses*: Image Processing, Signal Compression, Optimal Estimation and Filtering

B.Tech, Electronics and Communication Engineering, National Institute of Technology 2016-2020

- *Awards*: DAAD WISE Scholarship, MITACS Globalink Research Intern Award
- *Relevant Courses*: Soft Computing, Antenna and Propagation, Software Engineering

TECHNICAL SKILLS

- **Programming & Frameworks**: C++, C, Python, PyTorch
- **Software**: MATLAB, Inkscape

PUBLICATIONS

- **Monsij Biswal**, Tong Shao, Kenneth Rose, Peng Yin, Sean Mccarthy. "Steganerv: Video steganography using implicit neural representation." in IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). IEEE, 2024.
- **Monsij Biswal**, Kruthika Koratti Sivakumar, Ting-Lan Lin, Kenneth Rose. "Transform Domain Temporal Prediction for Dynamic Point Cloud Compression." in IEEE International Workshop on Multimedia Signal Processing (MMSP). IEEE, 2023.
- **Monsij Biswal**, Aniruddha Chandra, Aniq Ur Rahman et al. "On the characterization of beam misalignment in outdoor-to-indoor 60 GHz mmWave channel." in European Conference on Antennas and Propagation (EuCAP). IEEE, 2021.
- Michael Lutz, **Monsij Biswal**. "Supervised Noise Reduction for Clustering on Automotive 4D Radar." in IEEE Symposium Series on Computational Intelligence (SSCI). IEEE, 2021.
- Aniq Ur Rahman, **Monsij Biswal**. "Error-tolerant Beam Steering of mmWave Antennas by Trajectory Estimation of Highway Vehicles." in 11th International Conference on Communication Systems & Networks (COMSNETS). IEEE, 2019.

RESEARCH EXPERIENCE

Graduate Student Researcher, Signal Compression Lab, UCSB *Present*

- Working towards compressing VR180 stereoscopic videos with geodesic motion compensation.
- Experience in video coding using HEVC with 360Lib.

Research Scientist Intern, Amazon Prime Video, Seattle *June 2024 - September 2024*

- Improved the performance of an existing video post-processing approach for better rate-distortion performance.
- Developed subjective study interface for collecting subjective test results with 4K UHD HDR content.

ATG Imaging Research Intern, Dolby Laboratories, Sunnyvale *June 2023 - September 2023*

- Explored the emerging trend of implicit neural representations for video representation and compression.
- Researched green field ideas to integrate data hiding within deep neural models representing videos.

Summer Research Intern, Scientific Computing Group, UCSB *June 2022 - August 2022*

Project: On the exploration of training networks on tiles of satellite images

- Leveraged the performance of tile-based training to achieve near state-of-the-art results for semantic image segmentation.
- Investigated the effects of varying tile size, data augmentation and network architecture on training complexity and overall classification score.

Mentor, Research Mentorship Program, UCSB

June 2021 - August 2021

Project: Supervised Noise Reduction for Clustering on Automotive 4D Radar

- Mentored and collaborated with a junior researcher on the removal of noisy radar returns prior to clustering using density based algorithms.
- Utilized supervised learning to predict noisy points on 4D radar point clouds by leveraging historical data.
- Co-authored research article that has been published at IEEE SSCI 2021.

Research Intern, Christian Albrechts Universität zu Kiel, Germany

May 2019 - July 2019

Project: Custom Neural Network Training Function for Optical Transmission System

- Analyzed various neural network training algorithms for equalizing a higher order modulation format in an optical transmission system.
 - Implemented custom training function and presented research findings to the group.
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TEACHING EXPERIENCE (TA)

AY 2024-25

- ECE 5: Introduction to Electrical and Computer Engineering
- ECE 139: Probability and Statistics

AY 2023-24

- ECE 5: Introduction to Electrical and Computer Engineering
- ECE 139: Probability and Statistics

AY 2022-23

- ECE 3: Introduction to Electrical Engineering
- ECE 10B/BL: Fundamentals of Analog and Digital Circuits

AY 2021-22

- ECE 3: Introduction to Electrical Engineering
- CMPSC 111: Introduction to Computational Science
- ECE 139: Probability and Statistics
- PHYS 129L: Introduction to Scientific Computation

AY 2020-21

- ECE 15A: Fundamental of Logic Design
 - ECE 139: Probability and Statistics
 - CMPSC 24: Problem Solving with Computers II
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LEADERSHIP EXPERIENCE

Vice-President, ECE Graduate Student Association, UCSB

September 2021 - December 2024

- Lead a team of 4 graduate students by coordinating logistical tasks and delegating responsibilities.
- Facilitated group discussion and meetings to plan research seminars, lightning talks and social hours.

Co-chair, Beyond Academia, UCSB

August 2021 - March 2022

- Collaborated with multiple teams for organizing a career exploration conference comprising of 200+ attendees and 30+ presenters.
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Last updated : October 2025